Two-input motor boards (IN1 / IN2)

PSC Robotics: wiring and example code for DC motor driver boards with two inputs, such as the DRV8871. One board drives one motor.

Every motor board

Motor board pin	Connect to
VM / Power +	Motor battery + (at least 6.5V for a DRV8871, for example 12V)
GND / Power -	Motor battery -
OUT1, OUT2	The motor's two wires
IN1, IN2, GND	The Raspberry Pi or the Arduino: see the tables below

The motor battery's ground must be joined to the Raspberry Pi's or Arduino's ground. The GND connection in the tables below does this. Never power the motors from the Raspberry Pi or the Arduino. If a motor turns the wrong way, swap its two wires on OUT1 and OUT2.

Wired to the Raspberry Pi

Motor board pin	Left motor	Right motor
IN1	GPIO 20 (header pin 38)	GPIO 19 (header pin 35)
IN2	GPIO 21 (header pin 40)	GPIO 26 (header pin 37)
GND	GND (header pin 39)	GND (header pin 34)

Wired to the Arduino

Motor board pin	Left motor	Right motor
IN1	Pin 9	Pin 5
IN2	Pin 10	Pin 6
GND	Arduino GND	Arduino GND

IN1 and IN2 must both go to PWM pins. On an Arduino Uno these are 3, 5, 6, 9, 10 and 11 (marked ~). Needs version 6 or later of the PSC Arduino firmware.

Quick reference

```
from sr.robot3 import BRAKE, COAST, Robot

robot = Robot()

# Raspberry Pi pins (BCM GPIO numbers):
motor = robot.brain.dc_motor(20, 21)   # IN1, IN2
# ...or Arduino pins (both must be PWM pins):
# motor = robot.arduino.dc_motor(9, 10)       # IN1, IN2

motor.power = 0.5     # -1 (full reverse) to 1 (full forwards)
motor.power = BRAKE   # stop quickly
motor.power = COAST   # let it spin freely
```

To run an example, copy the code on the next pages into a file called robot.py at the top level of your USB stick, plug it into the robot and press the start button.

Both examples are also attached to this PDF, as robot-brain.py and robot-arduino.py. If copying loses the indentation, save the attachment instead (in most PDF viewers, open the attachments or paperclip panel), then rename it to robot.py.

Example: Raspberry Pi pins

robot.py (brain version)

```
"""
Drive two DC motors through two-input motor driver boards (IN1 and IN2), such as
the DRV8871, wired to the Raspberry Pi.

Wiring (BCM GPIO numbers, header pin in brackets). Use one driver board per motor.

    Left motor driver           Right motor driver
      IN1 -> GPIO 20 (pin 38)     IN1 -> GPIO 19 (pin 35)
      IN2 -> GPIO 21 (pin 40)     IN2 -> GPIO 26 (pin 37)
      GND -> GND     (pin 39)     GND -> GND     (pin 34)

    On each driver board:
      VM / power +  -> motor battery + (6.5V to 45V for a DRV8871, e.g. 12V)
      GND / power - -> motor battery -
      OUT1, OUT2    -> the motor's two wires

The motor battery's ground must be joined to the Raspberry Pi's ground (the GND
pins above do this through the driver board). Never power the motors from the Pi.

Copy this file to the top of your USB
stick, named robot.py, plug it into the robot and press the start button.
"""
from sr.robot3 import BRAKE, COAST, Robot

robot = Robot()  # Waits here until the start button is pressed

left_motor = robot.brain.dc_motor(20, 21)   # IN1, IN2
right_motor = robot.brain.dc_motor(19, 26)  # IN1, IN2


def drive(left, right, seconds):
    """Set both motors' power (-1 to 1) for a number of seconds."""
    left_motor.power = left
    right_motor.power = right
    robot.sleep(seconds)


print("Forwards at half speed")
drive(0.5, 0.5, 2)

print("Stop")
drive(BRAKE, BRAKE, 1)

print("Backwards at half speed")
drive(-0.5, -0.5, 2)

print("Turn on the spot to the right")
drive(0.5, -0.5, 1)

print("Turn on the spot to the left")
drive(-0.5, 0.5, 1)

print("Full speed forwards")
drive(1, 1, 1)

print("Coast to a stop")
drive(COAST, COAST, 1)

print("Done")
# The motors are also stopped automatically when your code ends.
```

Example: Arduino pins

robot.py (arduino version)

```
"""
Drive two DC motors through two-input motor driver boards (IN1 and IN2), such as
the DRV8871, wired to the Arduino.

Wiring (Arduino Uno pin numbers). Use one driver board per motor. IN1 and IN2
must both go to PWM pins: on an Uno these are 3, 5, 6, 9, 10 and 11 (marked ~).

    Left motor driver           Right motor driver
      IN1 -> pin 9                IN1 -> pin 5
      IN2 -> pin 10               IN2 -> pin 6
      GND -> Arduino GND          GND -> Arduino GND

    On each driver board:
      VM / power +  -> motor battery + (6.5V to 45V for a DRV8871, e.g. 12V)
      GND / power - -> motor battery -
      OUT1, OUT2    -> the motor's two wires

The motor battery's ground must be joined to the Arduino's ground (the GND
connections above do this). Never power the motors from the Arduino.

Needs version 6 or later of the PSC
Arduino firmware (download it from the Arduino page of the docs). Copy this file
to the top of your USB stick, named robot.py, and press the start button.
"""
from sr.robot3 import BRAKE, COAST, Robot

robot = Robot()  # Waits here until the start button is pressed

left_motor = robot.arduino.dc_motor(9, 10)  # IN1, IN2
right_motor = robot.arduino.dc_motor(5, 6)  # IN1, IN2


def drive(left, right, seconds):
    """Set both motors' power (-1 to 1) for a number of seconds."""
    left_motor.power = left
    right_motor.power = right
    robot.sleep(seconds)


print("Forwards at half speed")
drive(0.5, 0.5, 2)

print("Stop")
drive(BRAKE, BRAKE, 1)

print("Backwards at half speed")
drive(-0.5, -0.5, 2)

print("Turn on the spot to the right")
drive(0.5, -0.5, 1)

print("Turn on the spot to the left")
drive(-0.5, 0.5, 1)

print("Full speed forwards")
drive(1, 1, 1)

print("Coast to a stop")
drive(COAST, COAST, 1)

print("Done")
# The motors are also stopped automatically when your code ends.
```